Nitzan Raz

Software Architect

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EXPERIENCE

Independent Consulting - *Software Architect*

2025 - Present

- Proudest moments:
  - Modified complex multimedia ingestion pipeline (over 10K assets a day) from 5% success rate to 97%
  - Unified 5 different "management portals" of questionable quality to a single interface, actively maintained by the team
  - Saved $5K a month by migrating a network-intensive service to a new cloud, zero downtime
  - Unified onprem, BYOC, SaaS deployment strategies into a single k8s-based approach
- Working with leadership
  - Enabled engineering teams (ranging from 10-person startups to near-enterprise scale) to focus on product and core competencies by owning technical capabilities outside their expertise
  - Collaborated with CTOs on strategic technical planning, rapidly prototyping different approaches to validate architectural decisions before full investment
- Areas of focus:
  Software development (refactoring, rewriting, consolidation), infrastructure operations, data reliability, ML integration, security posture, and developer experience

UGLabs, Tel Aviv - *Software Architect*

2024 - 2025

- Designed and implemented tech stack
  - Terraform, k8s, Docker Images, GH Actions + ArgoCD for CI/CD
    Previously was hand-deployed Google Run
  - Multi-env, IaC, highly secure, DRP
  - Multi-Cloud with secure cross-communication
  - Included GPU-bound LLM services (realtime inference)
  - Proprietary data storage scheme compliant with COPPA, GDPR (audited, custom encryption, least privilege, lifecycle), alongside ML training pipeline for continuous self improvement.
- Capacity and throughput planning, budgeting and testing
- Deployed end to end (Unity to microservices to SQL) distributed monitoring used for troubleshooting and performance tuning
- SME on architecture for collaboration talks with providers and partners
- Special projects including IoT, synthetic transactions, audio compression

AngelSense, Tel Aviv - *Software Architect*

2023 - 2024

- Proudest moments:
    - Shrinking SMS costs to 3% by moving to SMPP
    - Reducing build time from 30 minutes to 3 minutes with big ephemeral builders
    - Improved build success rate from ~70% to 97% by eliminating moving parts (Nexus, unit tests with external dependencies) and introducing per-commit testing
    - Planning and implementing IaC with Terraform, Jcasc, EKS
    - Migrating noSQL storage from manually provisioned Cassandra to DynamoDB, no downtime. Planned and executed infra and coding
    - Moving codebase from Java8, Spring Boot 2.1 to Java17, Spring Boot 2.7. No rollback or downtime
    - Designing and PoCing strategic product features
- Working with Leadership
    - Letting VP R&D focus on owning the dev/product flow by taking over ops
    - Involved in prioritization and implementation of strategic software features
- Special missions
    - Working on features outside team expertise, delivering PoCs for later expansion by developers
    - Improving developer quality of life by improving the codebase and eliminating techdebt
- Firefighting
    - Driving the process - identifying, remediating, and post-mortem
    - Communicating with non-tech stakeholders

Facebook, Tel Aviv - *Software Engineer*

2016 - 2022

- Diverse problem finding and solving (see below), focusing on reliability, performance, scale and security (as opposed to new features).
- Worked with both user-facing teams and internal teams.

Proudest moments:

- Unblocked a company-wide public launch, 2 days after joining the team.
  User Care discovered an encryption gap in their kubernetes logging flow that has to be solved by end of day. Solution involved Linux STDOUT hijacking and an uncommon encryption scheme.
- Rewrote a week-over-week engagement monitoring system to unblock a big migration of business logic outside of the main server binary, making those servers unable to report on user engagement.
  I moved the reporting mechanism upstream by using a new data platform that encompassed all of the engagement data, and adapted the platform to support real time monitoring (rather than daily-resolution aggregations).
- Migrated the billion-user-serving infrastructure of Facebook Lite from a few big servers to many tiny servers.
  Process was nontrivial as the executable was Java with a very large in-heap cache, and running it unmodified reduced user capacity dramatically.
- Crucial part of the initial response to the 2019 plaintext-password incident.

- Saved 760 monthly hours of server time in 6 LoC diff.
  Colleague wanted to rewrite the entire e2e testing system as it was taking “too long” to test every commit, Rewrite was expected to take 2 months.
  I instead moved the client and server builds to run in parallel, and the rewrite plan became unnecessary, allowing the team to invest their time in more impactful projects.
- Wrote and fully delivered the “mark notification as read” feature on Facebook’s website.

Dynamic Yield, Tel Aviv - *DevOps*

2014 - 2016

- First DevOp in startup, moved infrastructure from manual to automated
- Configuration management (Chef), IaC, central logging, auto-scaling
- CI/CD automation and deployment pipelines
- Chef contributor (SELinux, Shellout)
- Security, monitoring, and big data operations

Mamtam (Israeli Navy) - *DevOps / Team Lead*

2008 - 2014

- Sharepoint / DotNet / MsSQL DevOps
- Active Directory / Exchange / SCCM / HyperV admin
- Led a team of 9
- Commended for excellence

EDUCATION

Open University, Israel - *Bachelor of Computer Science (**GPA 93**)*

2010 - 2016